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Sensitivity of Escarpment Evolution to Lithology and Climate Forcing:

Forward Landscape Evolution Study from the Swabian Alb

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Core message

Across 1,728 forward simulations, high-magnitude erosion is not randomly distributed. It remains concentrated along the Albtrauf escarpment, while the plateau interior stays comparatively stable. Lithology controls the spatial pattern; climate mainly amplifies incision where channels are already active.

1 Myr	1,728	250 m	8
simulation horizon	model ensemble	LEM grid	lithological units

1 Why this matters

- Long-term surface stability depends on whether erosion remains localized or migrates into plateau interiors.
- The Swabian Alb is a cuesta landscape with an active Albtrauf escarpment next to a low-relief karstified plateau.
- Key question: does future erosion remain focused along the Albtrauf escarpment under uncertain boundary conditions?

2 Numerical modelling framework

Boundary data:
DEM 250 m initial surface
3-D lithology: 8 units
spatial uplift field
CHELSA precipitation
karst + weak infill

Landlab engine:
LossyFlowAccumulator D8
FastscapeEroder
LinearDiffuser
Lithology update
open outflow boundaries

Time loop:
uplift
scale precipitation
route flow + karst loss
incise + diffuse
update exposed lithology

Model outputs:
cumulative erosion
topographic change
surface lithology
discharge loss
hotspots + profiles + divide shift

$$\partial z/\partial t = U - K A^m S^n + k_d \nabla^2 z$$

Erodibility K varies by lithological unit and updates as erosion exposes deeper strata.

Model setup: 1 Myr simulation | 10 kyr intervals | $m = 0.45$, $n = 1$ | outputs saved every 100 kyr

Baseline Scenario:
Upref | Rock2 refined K field | present precipitation | 30 m infill | 60% karst loss | m = 0.45 , n = 1

Rock2 = refined lithology-dependent K field within uncertainty bounds.

3 Validation and limitations

Validation:

- Concavity θ calibrated from 74 Neckar tributaries: mean θ ≈ *0.44*.
- K constrained using cosmogenic erosion rates and channel steepness.
- Baseline compared with published erosion and escarpment-retreat patterns.
- Ensemble tests pattern robustness under parameter uncertainty.

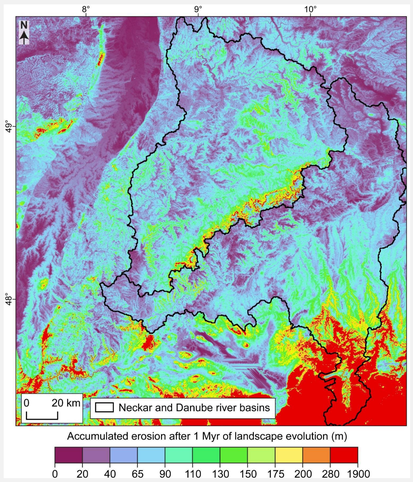
limitations:

- Robust spatial patterns, not exact future topography.
- Karst simplified as runoff loss.
- No landsliding or nonlinear hillslope failure.
- 250 m grid resolves regional patterns, not site-scale hazards.

Important: Limits affect exact magnitudes more than the escarpment-versus-plateau pattern.

4 Main result: erosion hotspots remain focused along the Albtrauf

Baseline 1 Myr erosion hotspots show a continuous Albtrauf belt, while the plateau interior remains sparse and discontinuous.



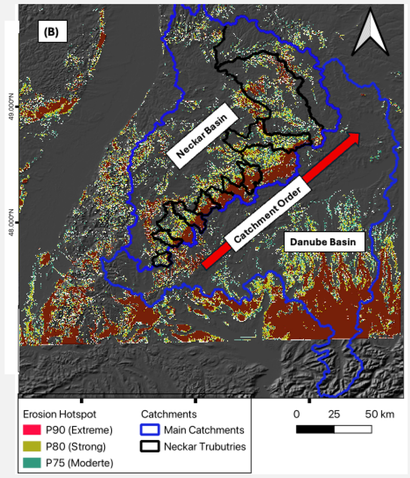
Hotspots are grid cells where cumulative erosion after 1 Myr exceeds domain-specific erosion percentiles: **P_{75} = upper 25%, P_{80} = upper 20%, P_{90} = upper 10%** of erosion values.

78.7% Albtrauf above P_{75}

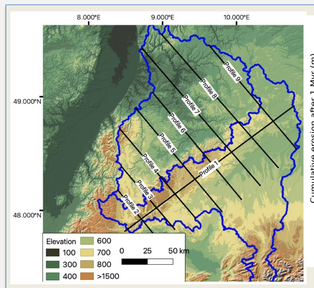
57.7% Albtrauf extreme P_{90}

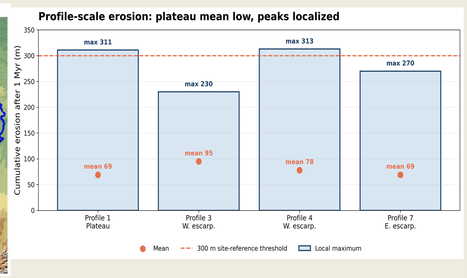
14.1% Plateau above P_{75}

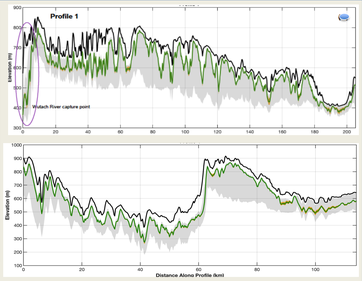
Negligible Plateau above P_{90}



The plateau is not erosion-free, but it remains consistently less affected than the escarpment across the tested scenarios.

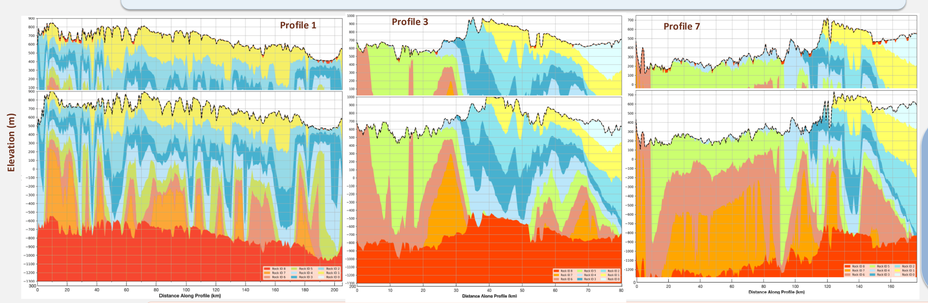






Mean erosion remains low along plateau-scale profiles, but local maxima exceed 300 m where profiles intersect active escarpment or tributary zones.

Stratigraphic feedback: incision exposes weaker units near the escarpment



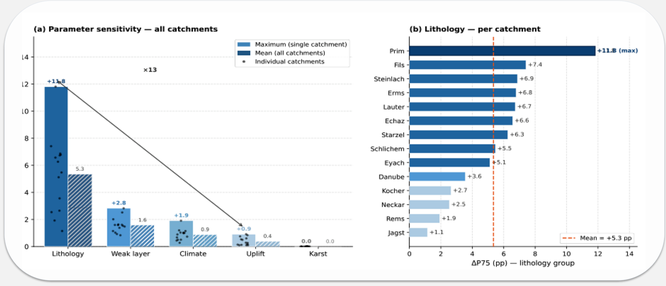
Plateau architecture remains largely intact; the strongest lithological reorganization is confined to the escarpment front.

Rock ID: 0 weak infill | 1 Pre-Triassic | 2 Lower Triassic | 3 Middle Triassic | 4 Upper Triassic | 5 Lower Jurassic | 6 Middle Jurassic | 7 Upper Jurassic–Cretaceous | 8 Post-Cretaceous/alluvium

Mechanism: rock strength contrast + drainage asymmetry + exposure of weaker layers.

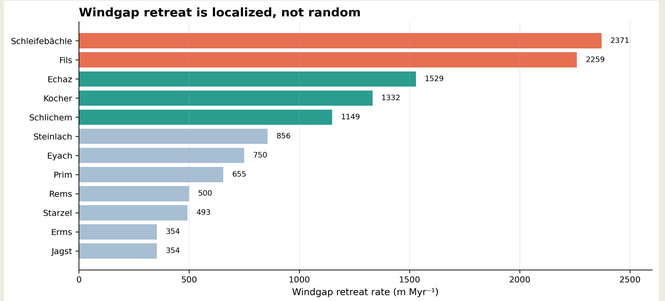
5 Lithology controls the sensitivity pattern

Lithology produces the largest change in hotspot coverage. Climate increases incision mainly in already active channels, while uplift and karst have limited influence on the spatial pattern.



6 Divide migration is localized near active tributaries

Divide migration is strongest where high-erosion hotspots are already concentrated, linking drainage reorganization to focused Albtrauf incision.



7 Conclusion

1. High-magnitude erosion remains concentrated along the Albtrauf.
2. Plateau interiors remain comparatively stable across tested 1 Myr scenarios.
3. Lithology controls where erosion persists; climate amplifies already active channels.
4. The model supports regional screening, not site-scale prediction.

8 Future work

- Couple surface erosion with subsurface hydrogeology and stratigraphic exposure.
- Test glacial–interglacial climate forcing instead of simplified precipitation scaling.
- Add landslide and nonlinear hillslope processes at escarpment scale.
- Compare modelled divide migration with mapped windgaps and river-capture indicators.

Selected references: Whipple & Tucker (1999); Braun & Willett (2013); Barnhart et al. (2020); Karger et al. (2017); Steinig et al. (2024); Schaller et al. (2025); BGE project report (2025).

Figures/data derived from Sahil Kumar MSc thesis.